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NUMERICAL SIMULATION OF LOCAL HEAT TRANSFER AND SCALING OF A SYNTHETIC IMPINGING JET IN A CANONICAL GEOMETRY

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ABSTRACT

Synthetic jets are generated by an equivalent inflow and outflow of fluid into a system. Even though such a jet creates no net mass flux, net positive momentum can be produced because the outflow momentum during the first half of the cycle is contained primarily in a vigorous vortex pair created at the orifice edges whereas in the backstroke, the backflow momentum is weaker, despite the fact that mass is conserved. As a consequence of this, the approach can be potentially utilized for the impingement of a cooling fluid over a heated surface. In the present study, a canonical geometry is presented, in order to study the flow and heat transfer of a purely oscillatory jet that is not influenced by the manner in which it is produced. The unsteady Navier-Stokes equations and the convection-diffusion equation were solved using a fully unsteady, two-dimensional finite volume approach in order to capture the complex time dependent flow field. A detailed analysis was performed on the correlation between the complex velocity field and the observed wall heat transfer. A fundamental frequency, in addition to the jet forcing frequency, was found, and was attributed to the coalescence of consecutive vortex pairs. In some instances, this vortex pairing can lead to zones of low heat transfer. Two point correlations showed that the Nusselt number Nu , showed stronger correlation with the vertical velocity v although the spatial-temporal dependencies are not yet fully understood. It was found that the Reynolds number and the Strouhal number, are sufficient to successfully scale the problem at larger dimensions and this is presently being exploited in order to design validation experiments using jets large enough to allow careful local measurements.

INTRODUCTION

Synthetic jets can be created by periodically ejecting and ingesting fluid from or to an orifice. One of their remarkable features is that, even though they produce zero net mass flux per cycle, they generate positive net momentum, due to the difference in the dynamics of the fluid between the first part of the cycle (forward stroke) compared to the second part (back

stroke). For this reason, this phenomenon can be employed as a cooling technique by impinging the periodic fluid flow onto a heated surface.

The formation and evolution of a slit synthetic jet was investigated by Smith and Glezer [1]. When the ejection of fluid from the orifice takes place, a pair of vortices is generated at the exit edges due to flow separation. Subsequently, and impelled by their own positive momentum, this vortex pair advects away fast enough to just be slightly affected by the suction part of the cycle. In fact, during the back stroke the fluid flow enters into the orifice from the sides. The stroke length L_0 was defined as a characteristic length of the phenomenon associated with the amount of fluid moved during the forward stroke. It was shown that the instantaneous streamwise velocity, at a given downstream location, periodically reaches a maximum associated with the passage of the vortex pair.

Synthetic and steady jets have been compared experimentally for matched Reynolds numbers within the range $695 < Re < 2200$ [2]. The normalized velocity profiles collapsed for both cases at the downstream location where the centerline velocity is approximately 50% of the characteristic velocity. Similar to continuous jets, the synthetic jet width grows linearly with respect to the downstream direction, although it spreads more rapidly than the continuous case. The volume flux is also larger than the steady case, since more fluid is entrained by the vortex pair that rolls along with the flow.

In some cases, a merging of contiguous vortex pairs has been observed [3]. The power spectrum of the instantaneous velocity near the jet exit showed the appearance of sub-harmonics, associated with this coalescence of vortices. Such phenomenon had been also detected by Petersen [4].

The impingement of a synthetic jet over a heated surface has been studied by several authors [5-16]. An optimum jet-to-surface distance has been identified, where the heat transfer was maximized [5, 7, 8, 10, 11, 13]. When that distance is located within the so-called intermediate-flow regime, a more dynamic, and therefore more effective, mixing with colder ambient air augments local heat transfer rates. It was also observed experimentally that heat transfer can be enhanced by

operating the jet device producing near the resonant frequency of the cavity [5, 7-9].

The effect of the jet driving frequency and Reynolds number upon the heat transfer was examined by Pavlova and Amitay [8]. It was concluded that synthetic jets improved the heat removal by up to three times compared to steady jets. Moreover, they found that higher frequencies augment the heat transfer when the wall is close to the jet orifice.

Valiorgue et al. [11] identified two different flow regimes, for a fixed jet-to-surface distance in a round jet. For large stroke lengths at $L/D > 5$, a time-averaged recirculating vortex is established at $r/D = 2$. Since the distance traveled by a propagating vortex scales with the stroke length, it was proposed that the ratio L_0/H may determine the flow regime. In addition, a power law relationship between Nu and Re was presented. In a similar way, Chaudhari et al. [12], used experimental data to correlate Nu with Re , Prandtl number, jet-to-surface distance, jet diameter and heated area.

Recently, numerical analysis of the heat transfer phenomenon has been performed [13, 14, 16, 17]. Utturkar et al. [13] achieved fairly good agreement between numerical and experimental data. Laminar flow throughout was assumed, due to the high effective viscosity that dampen the small vorticity scale near the jet orifice.

Wang et al. [17] performed the numerical simulation of a cross flow synthetic jet over a heated zone. The vibration of the diaphragm used to produce the pulsating flow was simulated as a sine wave velocity function at the inlet boundary. Large eddy simulation was utilized and showed acceptable agreement for velocities and temperature distributions, compared to experiments.

Jagannatha et al. [14] also studied the problem numerically, assuming turbulent regime and utilizing the $k-\omega$ model. The jet was produced by a positive-negative volume change of a square cavity, where the lower wall was set to be oscillatory, similar to a diaphragm. Heat transfer was improved with respect to the steady jet by about 30%.

A numerical and experimental study of the cavity dimension effects upon a synthetic jet produced by a piezo-electric device was performed by Mane et al. [18]. Good agreement between both techniques was found, and disparities were attributed to the pre-stressed shape of the actuator and the pressure developed inside the cavity that loads the device, changing its outline with respect of the analytic profile proposed for the numerical simulation. They concluded that, in order to properly optimize the synthetic jet, it is necessary to have sufficient knowledge of the specific piezo-electric device that is being used. Chaudhari et al. [15] examined the effects on the heat transfer of synthetic jets with square, rectangular and slit orifice shapes. Rectangular orifices with larger hydraulic diameters and smaller aspect ratio were more favorable for space constrained systems.

MOTIVATION

Generally, in the previous work reported on the subject, the synthetic jets have been created using an oscillatory diaphragm,

which is actuated using piezo-resistive devices. One of the objectives of the present work is to de-couple the performance of the jet from the manner in which it is pumped into the system. If successful, the resulting problem will allow the investigation of fundamental behavior that is not influenced by the design of the device. Hence, a canonical problem is proposed where the oscillating jet is created in a frictionless 2-dimensional channel formed by two parallel plates. The fluid flow is pumped following a sinusoidal function of time and the driving frequency. A finite volume numerical method was used to perform numerical experiments on the canonical synthetic jet impinging over a heated surface. The numerical simulations were used to investigate (i) the evolution of the velocity and temperature fields throughout the process, (ii) the correlation between the heat transfer and the fluid dynamics, and (iii) the possibility of scaling the problem for larger dimensions.

NOMENCLATURE

c_p	specific heat, J/kg K
f	jet driving frequency, Hz
f_c	second fundamental frequency, Hz
h	heat transfer coefficient, W/m ² K
H	jet-to-surface distance, m
k	thermal conductivity, W/m K
L_0	stroke length, m
Nu	Nusselt number
p	pressure, Pa
q''	heat flux per unit area, W/m ²
Re	global Reynolds number
Re_u	Reynolds number based on u
Re_v	Reynolds number based on v
St	Strouhal number
t	time, s
T	period $1/f$, s
U_0	characteristic velocity of the jet, m/s
v_{in}	inlet velocity, m/s
v_{max}	inlet maximum velocity, m/s
$\mathbf{v}$	velocity vector, m/s
w	channel width, m
x, y	horizontal and vertical position, m

Greek symbols

θ	temperature, K
θ_{wall}	wall temperature, K
θ_∞	ambient temperature, K
μ	dynamic viscosity, Pa s
ν	kinematic viscosity, m ² /s
ρ	density, kg/m ³
φ	phase angle
ϕ	transport variable: u , v or θ

Subscripts

avg	time averaged
in	inlet condition
n	n th term

w_{wall} related to wall
 ∞ ambient condition

PHYSICAL SITUATION AND MATHEMATICAL MODEL

A canonical geometry was created to study the impingement of a synthetic jet over a heated wall. The fluid was considered to flow between two inviscid, adiabatic parallel plates separated by a distance w . The inlet flow was considered to be an oscillating uniform flow. A depiction of the canonical problem is shown in Fig. 1.

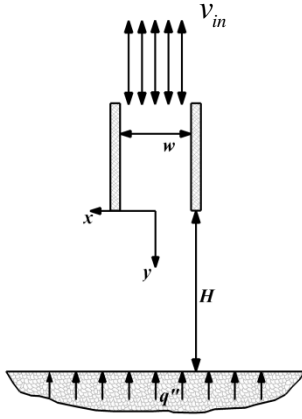


Fig. 1 Schematic of the physical domain that represents the canonical geometry.

The numerical formulation was subject to the following conditions: (i) Two-dimensional flow, (ii) unsteady laminar flow throughout, (iii) negligible viscous dissipation, and (iv) thermophysical properties not dependent on temperature. With these approximations, the governing conservation equations for mass, momentum and thermal energy are given as follows:

Continuity:

$$\nabla \cdot \mathbf{v} = 0 \quad (1)$$

Equation of motion:

$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla p + \mu \nabla^2 \mathbf{v} \quad (2)$$

Conservation of energy:

$$\rho c_p \frac{D\theta}{Dt} = k \nabla^2 \theta \quad (3)$$

The oscillating uniform velocity at the channel inlet obeys a sinusoidal function as follows:

$$v_{in}(t) = v_{max} \sin(2\pi f \cdot t) \quad (4)$$

The stroke length L_0 is the quantity that characterizes the amount of fluid ejected during the blowing part of the cycle

($0 < t/T < 0.5$), which can also be utilized to define the characteristic velocity U_0 of the jet.

$$L_0 = \int_0^{1/2f} v_{in}(t) dt \quad U_0 = L_0 f \quad (5)$$

Following the practice generally accepted in the literature, the global Reynolds number Re is defined in terms of the characteristic velocity U_0 and the jet width w . Such an approach has been used successfully to compare continuous and synthetic jets [2].

$$Re = \frac{U_0 w}{\nu} \quad (6)$$

The parameter that characterizes the periodic nature of the phenomenon is the Strouhal number St , which may also be defined as the inverse dimensionless stroke length.

$$St = \frac{fw}{U_0} = \frac{w}{L_0} \quad (7)$$

The pulsating flow impinges over a uniform heat flux wall located at a distance H from the channel exit thereby causing convective heat transfer from the wall to the jet. The heat transfer coefficient h and the Nusselt number Nu are defined as:

$$h = \frac{q''}{\theta_{wall} - \theta_{\infty}} \quad Nu = \frac{h \cdot w}{k} \quad (8)$$

The jet-to-surface distance was set as $H/w = 2.5, 5, 10, 17.5, 25$. Nu was monitored at 10 different locations along the wall: $x/w = 0, 0.5, 1, 1.5, 2, 3, 4, 6, 8, 10$. The horizontal and vertical velocities, u and v respectively, were also measured at the same x/w locations, although at a distance $\delta = 3w$ from the wall, except for $H/w = 2.5$ where $\delta = 1.5w$.

SOLUTION PROCEDURE

The transient mathematical model was solved by the finite volume method, by means of the commercial software FLUENTTM. The SIMPLE algorithm was selected to couple velocity and pressure as it performs well for unsteady flows where the time steps are small, as in the present study [19]. First order interpolation was considered for the discretized equations of continuity and momentum, whereas the energy equation was interpolated via the QUICK method.

The boundary conditions utilized are depicted in Fig. 2, as well as the grid size density where the control volumes with smaller areas are associated to darker colors. In the present study a hybrid structured-unstructured mesh was used. The structured part was set within the range $-80 < x/w < 80$ and $0 < y/w < H/w$, as well as in the channel inner region. With this approach, proper convergence is ensured near the zones where the simulation generates larger gradients. The rest of the computational domain was discretized using an unstructured mesh, as can be seen in Fig. 2, where the mesh pattern is more scattered due to a somewhat irregular distribution of the node

area size. Inside this region only minor gradients are produced, therefore generating less numerical instability, which justifies the use of this approach that, ultimately, is utilized to reduce computational costs.

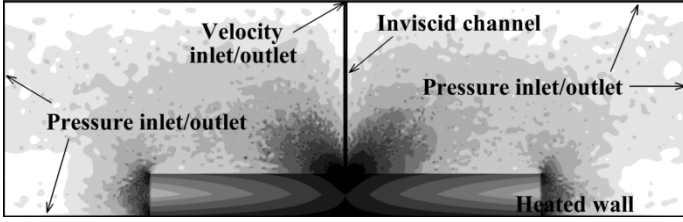


Fig. 2 Computational domain and grid size density.

Mesh size independence and dynamic time step

In order to prove mesh independence, four different mesh sizes were tested. The dependence of wall Nusselt number and jet mass flux on grid density is shown in Fig. 3. It can be seen that the mesh with 158,661 nodes introduced only minor deviations for the time averaged Nu , with respect to the finest grid (Fig. 3(a)), a tendency that is also observed in Fig. 3(b), where the two coarser grids showed observable deviations from the theoretical mass flux, leading to a computational error associated with the artificial generation of mass within the system.

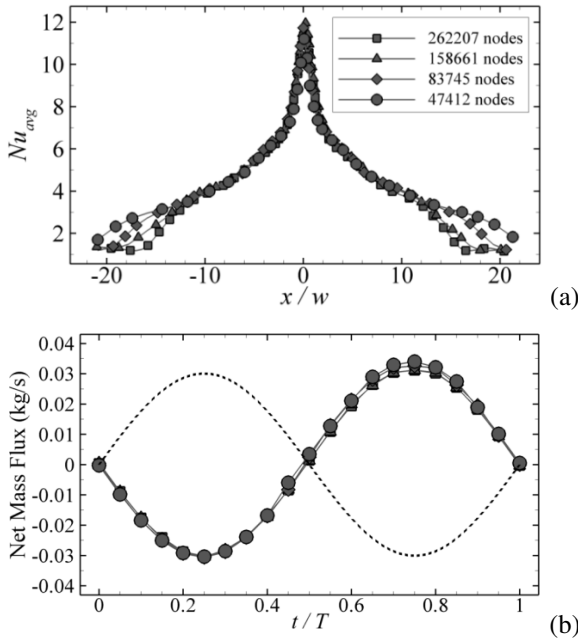


Fig. 3 (a) Time averaged Nusselt number distribution over the heated wall, and (b) Net mass flux at the pressure inlet-outlet boundary compared to inlet condition (dotted line).

Since one of the parameters changed in this study is the jet-to-surface distance H , the size of the computational domain, or the grid size, was varied as well. Thus, in order to maintain the grid independence, the density of nodes per unit length was kept constant in the vertical direction.

Three different time steps, defined as a function of the period T , were tested, as it is illustrated in Fig. 4. Even though the two cases almost coincided at every location along the heated wall, $T/100$ was the option elected, because this improves the convergence of the simulation, especially in the areas where the velocity gradients are high.

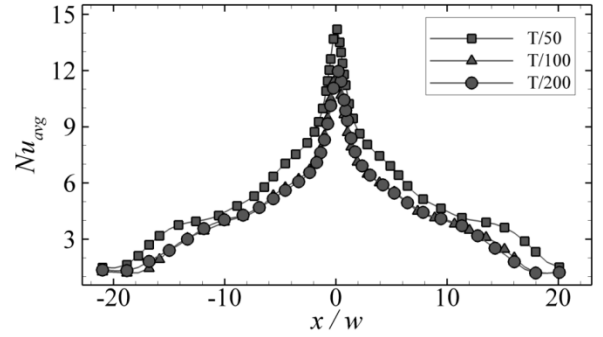


Fig. 4 Time averaged Nusselt number distribution over the heated wall for three different time steps.

RESULTS AND DISCUSSION

The Nusselt number evolution at $x/w = 0$, $H/w = 5$ and driving frequency $f = 800 \text{ Hz}$ is shown in Fig. 5 (a). The synthetic jet was generated by a periodic ejection-suction of mass, each cycle with a period $T = 1/f$. It can be observed that the frequency of the jet f was not the only fundamental frequency, suggesting that at least one other must exist. This is explained by the fact that T represents the period for the inlet/outlet mass flux, but does not represent the period of $Nu(t)$. Hence, in order to further study the nature of the problem, a spectral decomposition of the signal was performed, illustrated in Fig. 5 (b). As in previous studies [3, 4], the appearance of a lower frequency associated with coalescence of consecutive vortex pairs can be observed; it is the other fundamental frequency in the problem, f_c . Furthermore, the existence of harmonics that are multiples of f and f_c were detected.

These observations apply to all the cases covered in the present work, where f_c varied depending on f and H/w . Moreover, the ratio between the two fundamental frequencies remained constant $f/f_c = 2$ throughout. Therefore, if nf_c (where n is an integer > 1) represents the n th harmonic associated with f_c , then when n is even, the effect takes into account both fundamental frequencies. For this reason, this bi-periodic phenomenon (two fundamental frequencies), has a period $1/f_c$, which explains why, as it seen in Fig. 5 (a), Nu repeated every two cycles.

For deterministic systems, the instantaneous response of a variable at a given location x and y , may be represented as a sum of signals of the lowest frequency and its correspondent harmonics [20]. Thus for the present periodic jet, the Nusselt number and velocity components can be represented as:

$$Nu(t) = Nu_0 + \sum_{n=1}^{\infty} Nu_n \sin[2n\pi f_c t - (\varphi_{Nu})_n] \quad (9)$$

$$u(t) = u_0 + \sum_{n=1}^{\infty} u_n \sin[2n\pi f_c t - (\phi_u)_n] \quad (10)$$

$$v(t) = v_0 + \sum_{n=1}^{\infty} v_n \sin[2n\pi f_c t - (\phi_v)_n] \quad (11)$$

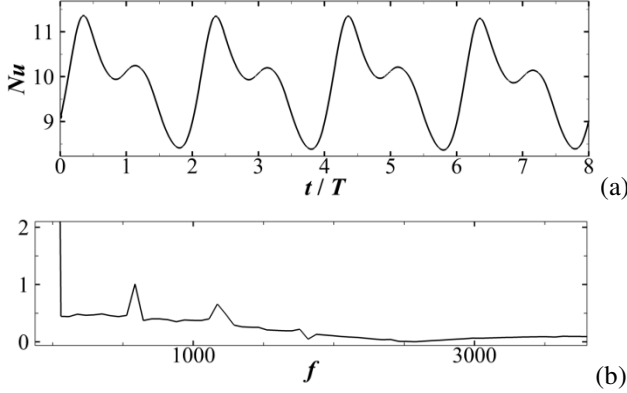


Fig. 5 (a) Nusselt number evolution at $x/w = 0$, $H/w = 5$, $St = 0.15$. (b) Spectral decomposition of Nu in (a).

The unknown coefficients of each function (amplitudes ϕ_n and phase angles $(\phi_\phi)_n$), were calculated by curve-fitting the numerical data. For practical purposes, only the first 7 terms were retained ($n = 1, 2 \dots 6$). This can be justified by the fact that, commonly (but not always, from a strict mathematical point of view), the amplitudes of higher order harmonics decrease exponentially with n . Indeed, it was generally observed in the present work that the terms corresponding to $n = 6$ and 7 showed amplitudes at least 3 orders of magnitude smaller than the first term.

If accurate, these expressions will permit proper identification of the steady part of the process, associated with the net positive momentum induced, from the unsteady part that is generated by the periodic manner in which the synthetic jet was produced. An example of how precisely this formulation may model the response of a given signal is shown in Fig. 6, for $H/w = 10$, $St = 0.15$ and $x/w = 1$. The symbols represent numerical data, whereas the dashed lines correspond to their fitted curve. It can be noticed that equation (9), as well as equation (11) represented non-dimensionally by $Re_v = vw/v$, could accurately capture the behavior of the phenomena. In Fig. 6, the fitted curve on Re_v allowed the phase shifting and therefore identification of the phase angle or time lag between the sensed v velocity and the response on the wall. This approach was not successful in all instances; therefore, in order to quantify the exactness of this method, the deviation of any of the curve-fitted theoretical functions $g_{th}(t)$ with respect to their correspondent numerical data $g_{num}(t)$ was defined as follows:

$$\varepsilon = \left| \frac{g_{th}(t) - g_{num}(t)}{[g_{num}(t)]_{avg}} \right| \cdot 100 \quad (12)$$

The mean deviation $\bar{\varepsilon}$ for all the cases studied, is illustrated in Fig. 7. The formulation proposed seemed to work well in general, however, it can be noticed that, for low H/w , $\bar{\varepsilon}$ tended to grow proportional to x/w and St . As the vortex pair travels away from the jet orifice, it loses momentum by dissipation of mechanical energy; therefore, the vortices diminish in coherence as they move farther from the centerline ($x = 0$). Additionally, St is inversely proportional to the stroke length, meaning that for larger St the characteristic length of the vortex is decreased, hence, the energy delivered to the vortex pair during the forward stroke, is reduced. These two combined effects explain why the phenomena are less dependent on their periodic nature, leading to the reduction in the accuracy of the fitting procedure.

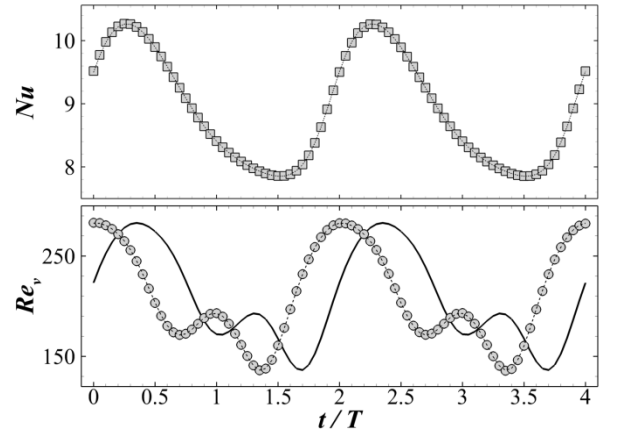


Fig. 6 Nu , and Re_v for $H/w = 10$, $St = 0.15$ and $x/w = 1$. Symbols represent numerical data, dashed lines their respective fitted curve and the solid line is the phase shifted Re_v .

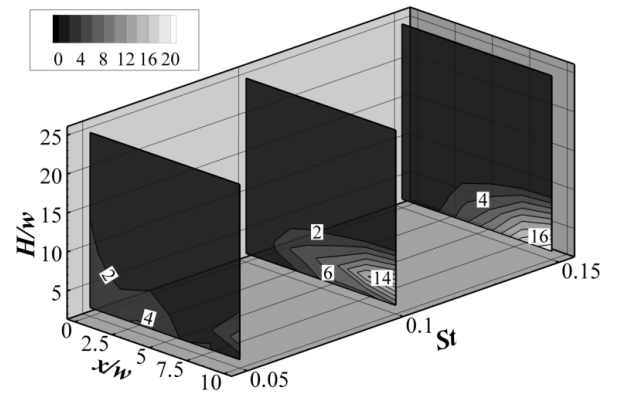


Fig. 7 Mean deviation of the fitted Nu (%) with respect of the numerical data, as a function of H/w and x/w for three different St .

The first three terms of equation (9), as well as the ratio Nu_1/Nu_2 are shown in Fig. 8. Nu_0 can also be interpreted as the time averaged Nusselt number, since it represents the steady part of the formulation, which is associated to the production of

positive net momentum by the synthetic jet. It can be observed from Fig. 8 (a) that there was practically no influence of St on Nu_0 , except at low H/w (≤ 5). As the wall was placed closer to the jet orifice, higher driving frequencies generated a more dynamic velocity field, with a subsequent improvement of the heat transfer related to a better mixing between the fluid closer to the wall with the colder ambient fluid. Moreover, at the smallest wall-to-orifice distance, a dipping in Nu_0 (at $x/w = 0$) was detected, an effect that increased with St . A lower stroke length, led to a weaker capacity of the vortex pair to properly energize the jet in order to penetrate into the stagnation zone.

Nu_0 generally decreased with increasing H/w and with x/w increasing, i.e. for increasing distance from the stagnation line. As the jet wall distance was increased, the jet impinged it with lower momentum, leading to lower heat transfer. Furthermore, farther from the centerline ($x/w = 0$), the wall jet thermal boundary layer increased in thickness, hence diminishing the wall jet's heat removal capacity. Closer to the centerline, a secondary optimum was observed in the area near $H/w = 10$, a spacing that has been favorable because of the previously mentioned fluid mixing induced by the vortex pair.

The variation of Nu_1 with respect to H/w and x/w is illustrated in Fig. 8 (b), for three different Strouhal numbers. This value is associated with the frequency related to vortex pairing f_c ; hence, its magnitude can be interpreted as the intensity of the effect of this vortex coalescence upon the heat

transfer. Nu_2 is depicted in Fig. 8 (c) in the same fashion as Nu_1 . This quantity is related to both, the forcing frequency f and the second harmonic of f_c .

In order to clarify the effect of f_c and f on the problem, the ratio Nu_1/Nu_2 is illustrated in Fig. 8 (d) for all the cases studied. This ratio increased with St and H/w , since, as previously mentioned the vortex size, decreases as St is augmented. This effect may generate the existence of more than one vortex pair in the region between the jet exit and the heated wall. In other words, as a new cycle is taking place, the previous vortex pair might have not reached the wall, which enhances the possibility of merging. By comparing Fig. 8 (a) and Fig. 8 (d), one can notice that, for $St = 0.1$ and 0.15 , zones of low Nu_1/Nu_2 are associated with areas of localized higher heat transfer (or higher Nu_0). This occurred near the stagnation zone ($x/w < 2$) for low H/w , and also for $H/w = 10$, which was where the secondary optimum was encountered. This suggests that the vortex pairing effect may worsen the heat transfer.

Nu , u and v were monitored at 10 different locations, where each variable shared the same x/w , although the control points for u and v were located at a distance δ from the wall, in order to obtain a non-zero signal. There is a time lag between the velocities and Nu , related to the length scale or distance between the monitoring locations of the velocities u , v in the fluid and the Nu on the wall.

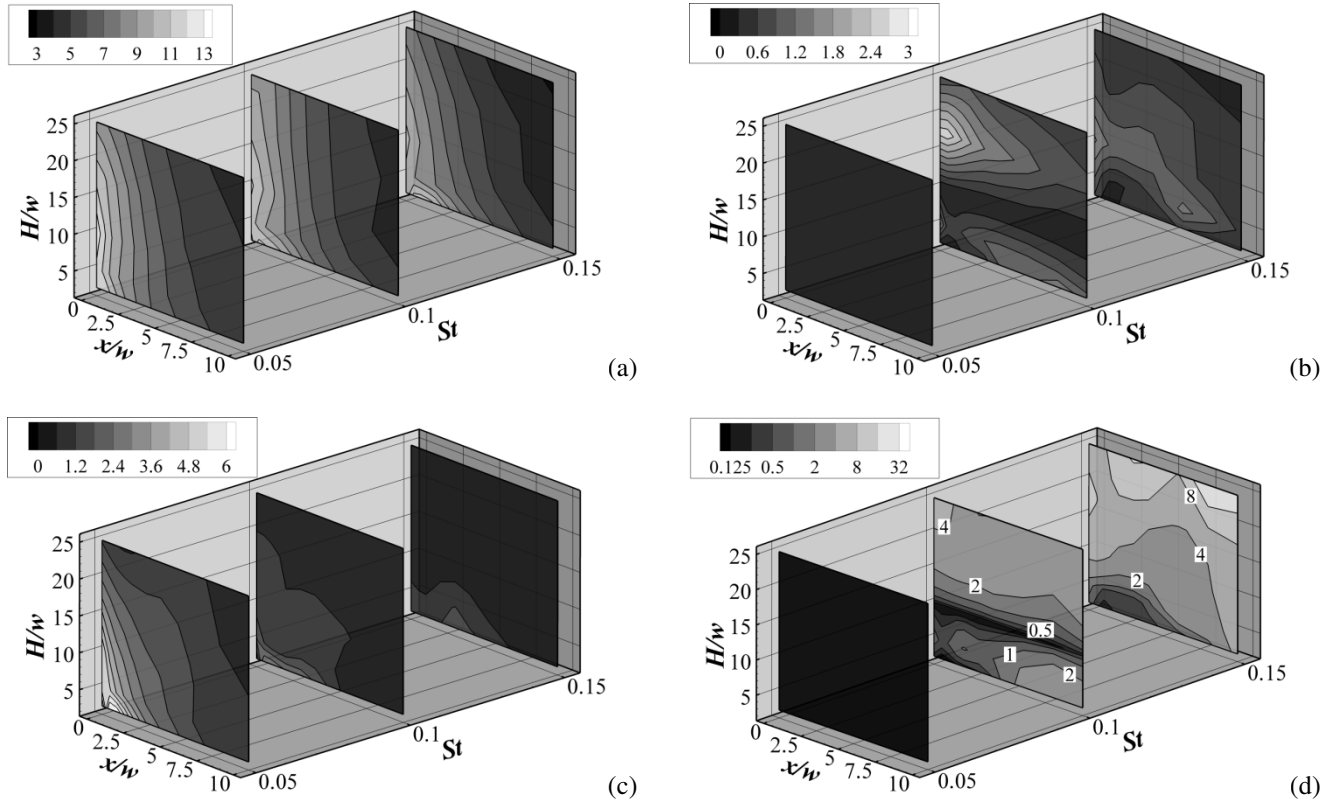


Fig. 8 Contour plots as a function of H/w and x/w for three different St : (a) Nu_0 , (b) Nu_1 , (c) Nu_2 , (d) ratio Nu_1/Nu_2 .

An additional advantage of the formulation proposed in equations (9) to (11) is that it permits the exact identification of the phase angle between each variable with respect of a given frequency. For instance, if $(\phi_{Nu})_1$ and $(\phi_v)_1$ represent the delay with respect of the moment where the vortex pairing took place, $(\phi_{Nu})_1 - (\phi_v)_1$ is the phase shifting between Nu and v , regardless of the moment when coalescence was generated or when the synthetic jet cycle started. Consequently, horizontal and vertical phase shifted velocities are defined, respectively, as follows:

$$u_{sh}(t) = u_0 + \sum_{n=1}^{\infty} u_n \sin \left\{ 2n\pi f_c t - \left[(\phi_u)_n + n((\phi_{Nu})_1 - (\phi_u)_1) \right] \right\} \quad (13)$$

$$v_{sh}(t) = v_0 + \sum_{n=1}^{\infty} v_n \sin \left\{ 2n\pi f_c t - \left[(\phi_v)_n + n((\phi_{Nu})_1 - (\phi_v)_1) \right] \right\} \quad (14)$$

An example of the phase shifting is depicted in Fig. 6. The solid line shows the phase shifted Re_v ; it can be observed that Re_v peaked simultaneously with Nu , therefore the time lag introduced by the time that it takes for a colder particle to reach the wall, has been successfully removed from the signal.

This technique allowed the proper computation of the phase corrected correlation of Nu with u and v .

The distribution of the correlation coefficient between Nu and u is showed in Fig. 9 (a), as well as between Nu and v (Fig. 9 (b)). Perhaps not surprisingly was correlated more strongly Nu with v , especially as St was increased.

Scaling of the Synthetic Jet

Very little is known about the dynamic scaling of the synthetic jet. We ask a simple but highly relevant question: What dimensionless groups fully capture the dynamic similitude of the jet? This has practical importance, because it is advantageous to experimentally study larger scale jets than smaller jets, due to instrumentation difficulties. The hypothesis is that Reynolds number Re and Strouhal number St are the dominant scaling parameters for the canonical jet geometry utilized in this study. Numerical experiments were conducted in order to investigate the hypothesis. Re was kept constant ($Re = 508$), varying $St = \{0.05, 0.1\}$, $H/w = \{5, 10, 17.5\}$ and the jet width $w = \{1 \text{ mm}, 2 \text{ mm}, 4 \text{ mm}\}$.

The instantaneous Nu and Re_v at two different H/w and two St are shown in Fig. 10. The different cases agreed well, for both fluid dynamics and the heat transfer at the wall. Some deviation of $w = 2 \text{ mm}$ and 4 mm with respect to the original problem ($w = 1 \text{ mm}$), can be observed in Fig. 10 (b). It can be noticed that $w = 2$ and 4 mm overlapped with each other, suggesting that the difference manifested might be related to numerical error, instead of physics associated with the problem itself. The latter may be explained by the fact that, for the method utilized, all the numerical parameters were kept constant regardless of w , which could have caused divergence since the computational domain was increased 2 and 4 times. Furthermore, commonly unsteady numerical simulations are sensitive to changes in the time step, which in this case varied

with the frequency as $\Delta t = 1/(100f)$, where f also needed to be changed in order to maintain St constant.

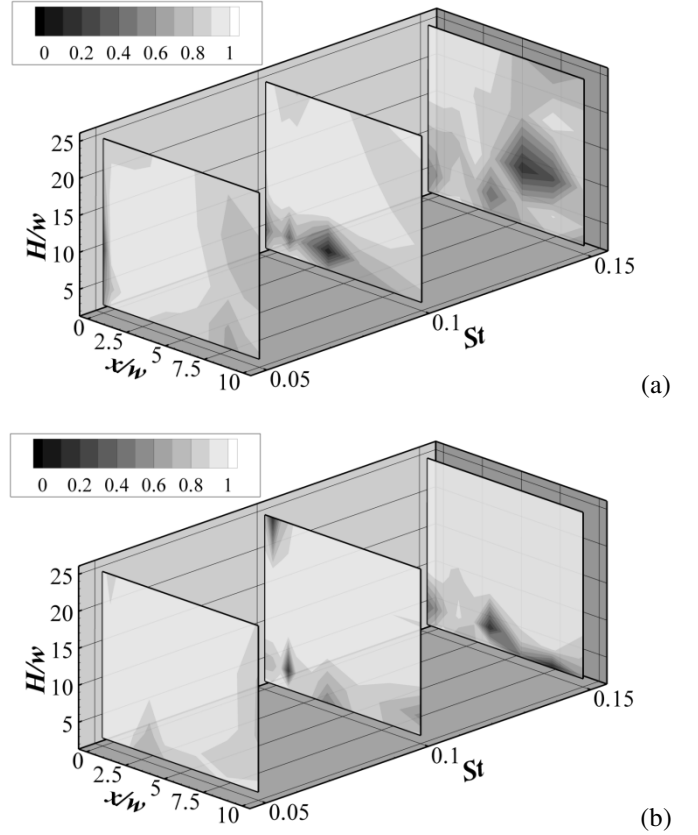


Fig. 9 Correlation coefficient between Nu and: (a) u , (b) v .

The deviation in % of $w = 2 \text{ mm}$ and 4 mm with respect to the original case was defined as:

$$e = \frac{\left| g(t) \Big|_{w=1\text{mm}} - g(t) \Big|_{w=\{2,4\}\text{mm}} \right|}{\left| [g(t)]_{\text{avg}} \Big|_{w=1\text{mm}} \right|} \quad (15)$$

The time averaged deviation e_{avg} is depicted in Fig. 11 for all the cases studied. Higher e_{avg} was obtained for $H/w = 17.5$ and $St = 0.05$ ranging from 3% to 7%. However, in general e_{avg} remained below 2%, showing that using Re and St as scaling parameters permitted the utilization of larger dimensions without altering the physical characteristics of the impinging synthetic jet.

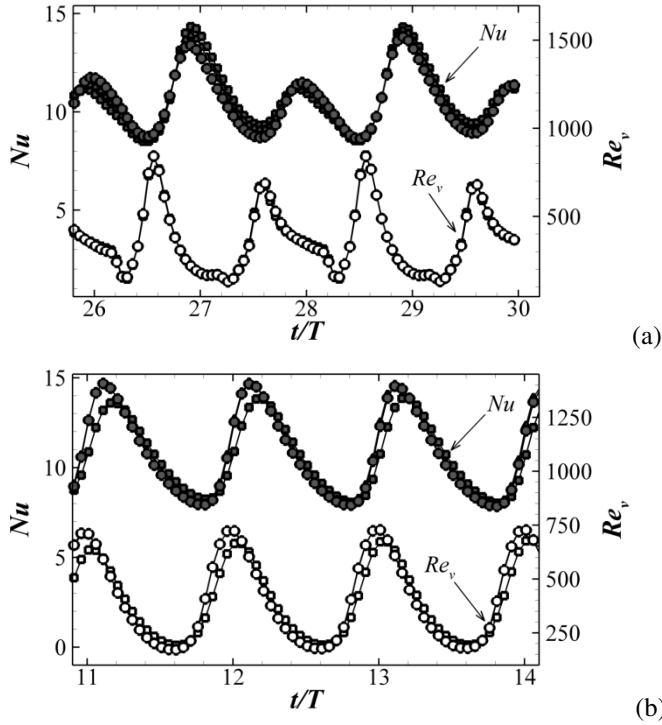


Fig. 10 (a) Nu and Re_v for $H/w = 5$, $St = 0.1$ and $x/w = 0$; (b) Nu and Re_v for $H/w = 17.5$, $St = 0.05$ and $x/w = 0$. Jet width $w = 1$ mm ($\square$), $w = 2$ mm ($\triangle$), $w = 4$ mm ($\circ$).

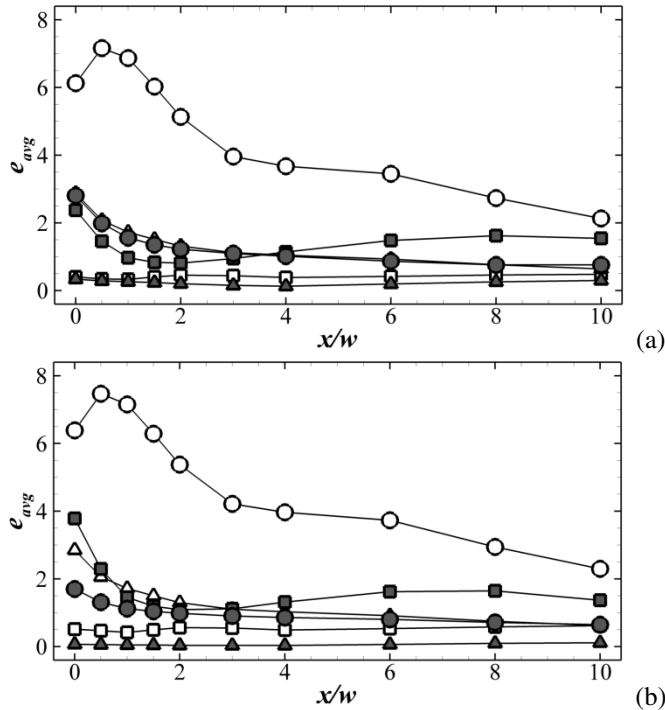


Fig. 11 Mean deviation from the case $w = 1$ mm for (a) $w = 2$ mm and (b) $w = 4$ mm. $H/w = \{5 (\square), 10 (\triangle), 17.5 (\circ)\}$. White symbols: $St = 0.05$, gray symbols: $St = 0.1$.

CONCLUSIONS

The numerical simulation of a canonical synthetic jet impinging on an evenly heated wall was presented.

The following general conclusions:

1. A second fundamental frequency is present in most cases and is associated with the coalescence of consecutive vortex pairs. Its effect was found to be directly proportional to the Strouhal number and H/w .
2. An analytical model was proposed to study the instantaneous response of Nu , u and v . This formulation allowed the decomposition of the response into its steady and unsteady parts. However, it tended to fail as the vortex coherence diminished, since the problem loses its periodic nature.
3. The time averaged Nu increased as H/w and x/w were reduced. In the vicinity of the stagnation zone, a secondary optimum was identified near $H/w = 10$, which was found to be the most favorable wall placement for both steady impingement and the fluid mixing created by the vortex pair.
4. The effect of the driving frequency f and the vortex pairing frequency f_c , was represented by the ratio Nu_1/Nu_2 , which increased with St and H/w . Some areas of localized higher heat transfer coincided with lower values of this ratio, suggesting that the vortex coalescence effect is unfavorable for the heat transfer.
5. By phase correcting the unsteady signals, an improved method to find their correlations was presented. It was shown that the heat transfer correlated more strongly with v rather than u .

A scaling analysis was performed using the Reynolds number Re and the Strouhal number St as the scaling parameters, for three different jet widths w . Good agreement was obtained between all the cases, showing that geometric scale-up may be possible for experimentally studying the impinging synthetic jet at larger physical dimensions.

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